

Fickle Trends (mode)

Giorgio enjoys doing endless statistical calculations in his spare time. Today, he collected the history of songs he ever played on his laptop, and started wondering about the many different music genres he followed during these last years.

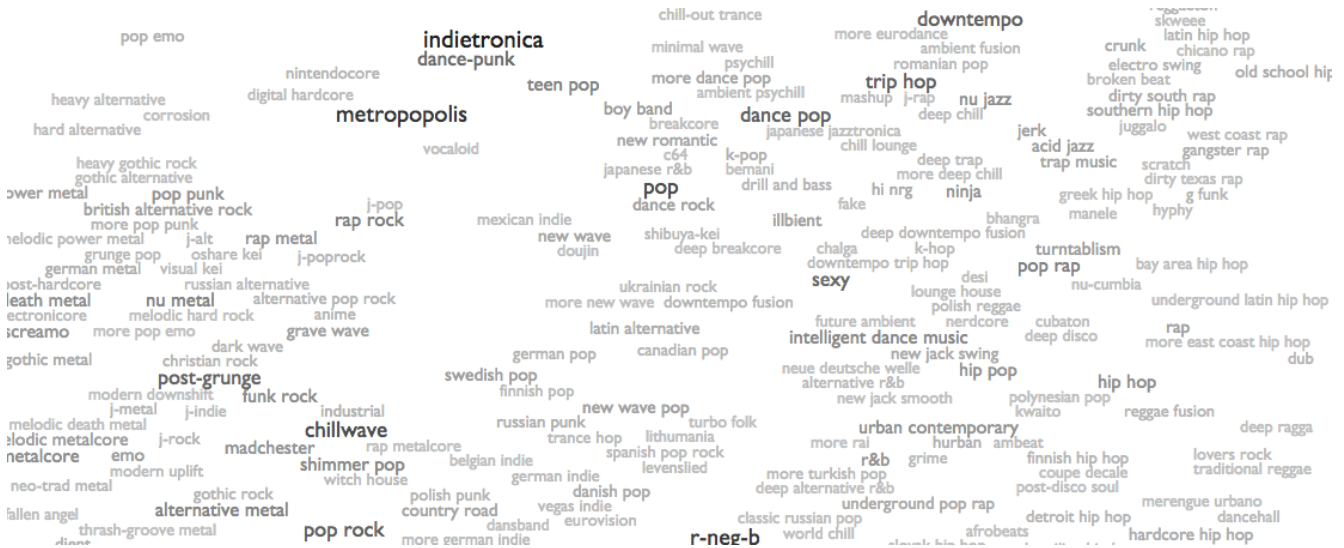


Figure 1: A spatial representation of music genres.

In particular, the list comprises N songs ordered chronologically, each of them tagged with a music genre G_i . Giorgio define as a *trend at time* $i = 1 \dots N$ any music genre which was a mode⁴ among the genres $G_0, \dots, G_{i-1}$ (that is, among the songs played before time i). How many different music genres were a trend at any past point in time?

👉 Among the attachments of this task you may find a template file `mode.*` with a sample incomplete implementation.

Input

The first line contains the only integer N . The second line contains N integers G_i .

Output

You need to write a single line with an integer: the number of different trends ever followed by Giorgio.

Constraints

- $1 \leq N \leq 100\,000$.
- $1 \leq G_i \leq 10^9$ for each $i = 0 \dots N - 1$.

⁴An element x is a mode in a list ℓ if it is one of the elements present most often in ℓ . A list can have multiple modes, for example, $\ell = [2, 0, 2, 1, 1]$ has modes 1 and 2.



Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1 [5 points]:** Examples.
- **Subtask 2 [35 points]:** $N, G_i \leq 100$ for all i .
- **Subtask 3 [40 points]:** $G_i \leq 1000$ for all i .
- **Subtask 4 [20 points]:** No additional limitations.

Examples

input.txt	output.txt
6 1 10 10 5 10 5	2
11 7 4 3 7 2 5 7 5 2 6 5	4

Explanation

In the **first sample case**, the trends are genres 1 and 10 which were both trending in time $i = 2$. Even though 5 is overall more frequent than 1, it has never been a trend.

In the **second sample case**, the trends are genres 3, 4, 7 (in time $i = 3$) and 5 (in time $i = 11$).